

Alignments

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Alignments

The core idea is that “to identical sequences correspond identical functions”

This assumption can be relaxed to state that the more two sequences are similar, bigger is the probability that they will share the same functions.

It should be clear the pivotal role played by sequences analyses in for the biological studies.

Alignments

Alignments have two primary use cases:

- Finding similar regions between two sequences - probing the relatedness of sequences.
- Finding which sequence (from many alternatives) is most similar to a query (input) sequence

Alignments

By “align” we means to put in columns the two (or more) sequences, residue after residue. Each of the columns will have a residue or a gap.

Taking the two sequences,

$s_1 = \text{AGATTCCAT}$ and $s_2 = \text{AGTCCCAT}$

a possible alignment would be:

A	G	A	T	T	C	C	A	T
A	G	—	T	C	C	C	A	T

Alignments

Given two generic strings, is there only one possible alignment?

$s_1 = \text{GATTACA}$, $s_2 = \text{GATCA}$

Alignments

Given two generic strings, is there only one possible alignment?

$s_1 = \text{GATTACA}$, $s_2 = \text{GATCA}$

G	A	T	T	A	C	A
G	A	T	-	-	C	A

G	A	T	T	A	C	A
G	A	-	T	-	C	A

G	A	T	T	A	C	A
			.			
G	A	T	C	A	-	-

Alignments

Is it possible to choose the “best” one?

$s_1 = \text{GATTACA}$, $s_2 = \text{GATCA}$

Alignments

Is it possible to choose the “best” one?

$s_1 = \text{GATTACA}$, $s_2 = \text{GATCA}$

G	A	T	T	A	C	A
G	A	T	—	—	C	A

G	A	T	T	A	C	A
G	A	—	T	—	C	A

G	A	T	T	A	C	A
			.			
G	A	T	C	A	—	—

Alignments

What rules the alignment process?

- Alignment scoring (parameters): the numerical values that tune the arrangement
- Alignment algorithm: global, local, semi-global, ...

Alignments and evolution

A	G	A	T	T	C	C	A	T
				.				
A	G	—	T	C	C	C	A	T

- conserved (MATCH)
- substituted (MISMATCH)
- insertion/deletion (INDEL)

Alignments and evolution

A	G	A	T	T	C	C	A	T
				.				
A	G	—	T	C	C	C	A	T

- conserved (MATCH)
- substituted (MISMATCH)
- insertion/deletion (INDEL)

It is possible to define an identity score to express the level of similarity of the two sequences according to the role played by evolution

Global and local alignment

GLOBAL we are looking for an arrangement that maximizes the similarities over the entire length of both sequences

LOCAL we are looking to identify the regions of maximal similarity between two sequences

Global alignment and Dynamic programming

Fundamental hypothesis:

Given two generic sequences s_1 and s_2 of length m and n and, a scoring function to maximize, given by the sum of the individual scorings of the alignment columns, it is possible to find the best alignment on top of

- the best alignment of the first $m - 1$ residues of s_1 with the first $n - 1$ residues of s_2
- the best alignment of the first m residues of s_1 with the first $n - 1$ residues of s_2
- the best alignment of the first $m - 1$ residues of s_1 with the first n residues of s_2

The scoring function

Given s_1, s_2 | $|s_1| = m, |s_2| = n$ the scoring function is defined on top of

- the best alignment of the first $m - 1$ residues of s_1 with the first $n - 1$ residues of s_2
- the best alignment of the first m residues of s_1 with the first $n - 1$ residues of s_2
- the best alignment of the first $m - 1$ residues of s_1 with the first n residues of s_2

The scoring function

Given $s_1, s_2 \mid |s_1| = m, |s_2| = n$ the scoring function is defined on top of

- the best alignment of the first $m - 1$ residues of s_1 with the first $n - 1$ residues of s_2
- the best alignment of the first m residues of s_1 with the first $n - 1$ residues of s_2
- the best alignment of the first $m - 1$ residues of s_1 with the first n residues of s_2

The scoring function

Given $s_1, s_2 \mid |s_1| = m, |s_2| = n$
the scoring function $a_{i,j}$ is defined on top of

↖ $a_{i-1,j-1}$

← $a_{i,j-1}$

↑ $a_{i-1,j}$

The scoring function

It is defined as:

$$a_{i,j} = \max \begin{cases} a_{i-1,j-1} + \sigma_{i,j} & \nwarrow \\ a_{i,j-1} + \text{gap} & \leftarrow \\ a_{i-1,j} + \text{gap} & \uparrow \end{cases}$$

where:

- $\sigma_{i,j} = \delta_{s_i^1, s_j^2}$ (MATCH, MISMATCH)
- $\text{gap} = 0$ is the penalty to align to a gap (INDEL)

The scoring function

In a realistic case the values would be:

$$a_{i,j} = \max \begin{cases} a_{i-1,j-1} + \sigma_{i,j} & \nwarrow \\ a_{i,j-1} + \text{gap} & \leftarrow \\ a_{i-1,j} + \text{gap} & \uparrow \end{cases}$$

- if $S_i^1 = S_j^2$ $\sigma_{i,j} = 5$ (MATCH) and $\sigma_{i,j} = -4$ otherwise (MISMATCH)
- $\text{gap} = -10$ INDEL

The algorithm

- build a matrix $A \in \mathcal{M}_{m+1,n+1}$ to store the alignment scores
- initialize the first row and column to 0
- compute the scores $a_{i,j}$ proceeding one row at a time, column by column
- traceback from $a_{m,n}$ to reconstruct the “optimal” alignment

Build the matrix

	-	A	G	A	T	T	C	C	A	T
-										
A										
G										
T										
C										
C										
C										
A										
T										

to store the score of the
partial alignments of s_1 and
 s_2

Initialize the matrix

	-	A	G	A	T	T	C	C	A	T
-	0	0	0	0	0	0	0	0	0	0
A	0									
G	0									
T	0									
C	0									
C	0									
C	0									
A	0									
T	0									

initializing the first row and column to 0 because they are the alignments with a gap.

Compute the partial alignments scores

	-	A	G	A	T	T	C	C	A	T
-	0	0	0	0	0	0	0	0	0	0
A	0									
G	0									
T	0									
C	0									
C	0									
C	0									
A	0									
T	0									

↖ $a_{i-1,j-1} + \sigma_{i,j} = 0 + \delta_{A,A} = 1$

Compute the partial alignments scores

	-	A	G	A	T	T	C	C	A	T
-	0	0	0	0	0	0	0	0	0	0
A	0									
G	0									
T	0									
C	0									
C	0									
C	0									
A	0									
T	0									

$$\nwarrow a_{i-1,j-1} + \sigma_{i,j} = 0 + \delta_{A,A} = 1$$

$$\leftarrow a_{i,j-1} = 0 + 0 = 0$$

Compute the partial alignments scores

	-	A	G	A	T	T	C	C	A	T
-	0	0	0	0	0	0	0	0	0	0
A	0									
G	0									
T	0									
C	0									
C	0									
C	0									
A	0									
T	0									

$$\nwarrow a_{i-1,j-1} + \sigma_{i,j} = 0 + \delta_{A,A} = 1$$

$$\leftarrow a_{i,j-1} = 0 + 0 = 0$$

$$\uparrow a_{i-1,j} = 0 + 0 = 0$$

Compute the partial alignments scores

	-	A	G	A	T	T	C	C	A	T
-	0	0	0	0	0	0	0	0	0	0
A	0									
G	0									
T	0									
C	0									
C	0									
C	0									
A	0									
T	0									

$$\nwarrow a_{i-1,j-1} + \sigma_{i,j} = 0 + \delta_{A,A} = 1$$

$$\leftarrow a_{i,j-1} = 0 + 0 = 0$$

$$\uparrow a_{i-1,j} = 0 + 0 = 0$$

1

Compute the partial alignments scores

	-	A	G	A	T	T	C	C	A	T
-	0	0	0	0	0	0	0	0	0	0
A	0	1								
G	0									
T	0									
C	0									
C	0									
C	0									
A	0									
T	0									

Compute the partial alignments scores

	-	A	G	A	T	T	C	C	A	T
-	0	0	0	0	0	0	0	0	0	0
A	0	1								
G	0									
T	0									
C	0									
C	0									
C	0									
A	0									
T	0									

$$\nwarrow a_{i-1,j-1} + \sigma_{i,j} = 0 + \delta_{A,G} = 0$$

Compute the partial alignments scores

	-	A	G	A	T	T	C	C	A	T
-	0	0	0	0	0	0	0	0	0	0
A	0	1								
G	0									
T	0									
C	0									
C	0									
C	0									
A	0									
T	0									

$$\nwarrow a_{i-1,j-1} + \sigma_{i,j} = 0 + \delta_{A,G} = 0$$

$$\leftarrow a_{i,j-1} = 1 + 0 = 1$$

Compute the partial alignments scores

	-	A	G	A	T	T	C	C	A	T
-	0	0	0	0	0	0	0	0	0	0
A	0	1								
G	0									
T	0									
C	0									
C	0									
C	0									
A	0									
T	0									

$$\nwarrow a_{i-1,j-1} + \sigma_{i,j} = 0 + \delta_{A,G} = 0$$

$$\leftarrow a_{i,j-1} = 1 + 0 = 1$$

$$\uparrow a_{i-1,j} = 0 + 0 = 0$$

Compute the partial alignments scores

	-	A	G	A	T	T	C	C	A	T
-	0	0	0	0	0	0	0	0	0	0
A	0	1								
G	0									
T	0									
C	0									
C	0									
C	0									
A	0									
T	0									

$$\nwarrow a_{i-1,j-1} + \sigma_{i,j} = 0 + \delta_{A,G} = 0$$

$$\leftarrow a_{i,j-1} = 1 + 0 = 1$$

$$\uparrow a_{i-1,j} = 0 + 0 = 0$$

1

Compute the partial alignments scores

	-	A	G	A	T	T	C	C	A	T
-	0	0	0	0	0	0	0	0	0	0
A	0	1	1							
G	0									
T	0									
C	0									
C	0									
C	0									
A	0									
T	0									

Compute the partial alignments scores

	-	A	G	A	T	T	C	C	A	T
-	0	0	0	0	0	0	0	0	0	0
A	0	1	1							
G	0									
T	0									
C	0									
C	0									
C	0									
A	0									
T	0									

$$\nwarrow a_{i-1,j-1} + \sigma_{i,j} = 0 + \delta_{A,A} = 1$$

Compute the partial alignments scores

	-	A	G	A	T	T	C	C	A	T
-	0	0	0	0	0	0	0	0	0	0
A	0	1	1							
G	0									
T	0									
C	0									
C	0									
C	0									
A	0									
T	0									

$$\nwarrow a_{i-1,j-1} + \sigma_{i,j} = 0 + \delta_{A,A} = 1$$

$$\leftarrow a_{i,j-1} = 1 + 0 = 1$$

Compute the partial alignments scores

	-	A	G	A	T	T	C	C	A	T
-	0	0	0	0	0	0	0	0	0	0
A	0	1	1							
G	0									
T	0									
C	0									
C	0									
C	0									
A	0									
T	0									

$$\nwarrow a_{i-1,j-1} + \sigma_{i,j} = 0 + \delta_{A,A} = 1$$

$$\leftarrow a_{i,j-1} = 1 + 0 = 1$$

$$\uparrow a_{i-1,j} = 0 + 0 = 0$$

Compute the partial alignments scores

	-	A	G	A	T	T	C	C	A	T
-	0	0	0	0	0	0	0	0	0	0
A	0	1	1							
G	0									
T	0									
C	0									
C	0									
C	0									
A	0									
T	0									

$$\nwarrow a_{i-1,j-1} + \sigma_{i,j} = 0 + \delta_{A,A} = 1$$

$$\leftarrow a_{i,j-1} = 1 + 0 = 1$$

$$\uparrow a_{i-1,j} = 0 + 0 = 0$$

1

Compute the partial alignments scores

	-	A	G	A	T	T	C	C	A	T
-	0	0	0	0	0	0	0	0	0	0
A	0	1	1	1						
G	0									
T	0									
C	0									
C	0									
C	0									
A	0									
T	0									

Compute the partial alignments scores

	-	A	G	A	T	T	C	C	A	T
-	0	0	0	0	0	0	0	0	0	0
A	0	1	1	1	1	1	1	1	1	1
G	0									
T	0									
C	0									
C	0									
C	0									
A	0									
T	0									

Compute the partial alignments scores

	-	A	G	A	T	T	C	C	A	T
-	0	0	0	0	0	0	0	0	0	0
A	0	1	1	1	1	1	1	1	1	1
G	0	1								
T	0									
C	0									
C	0									
C	0									
A	0									
T	0									

$$\nwarrow a_{i-1,j-1} + \sigma_{i,j} = 1 + \delta_{G,A} = 1$$

Compute the partial alignments scores

	-	A	G	A	T	T	C	C	A	T
-	0	0	0	0	0	0	0	0	0	0
A	0	1	1	1	1	1	1	1	1	1
G	0	1								
T	0									
C	0									
C	0									
C	0									
A	0									
T	0									

$$\nwarrow a_{i-1,j-1} + \sigma_{i,j} = 1 + \delta_{G,A} = 1$$

$$\leftarrow a_{i,j-1} = 0 + 0 = 0$$

Compute the partial alignments scores

	-	A	G	A	T	T	C	C	A	T
-	0	0	0	0	0	0	0	0	0	0
A	0	1	1	1	1	1	1	1	1	1
G	0	1								
T	0									
C	0									
C	0									
C	0									
A	0									
T	0									

$$\nwarrow a_{i-1,j-1} + \sigma_{i,j} = 1 + \delta_{G,A} = 1$$

$$\leftarrow a_{i,j-1} = 0 + 0 = 0$$

$$\uparrow a_{i-1,j} = 1 + 0 = 1$$

Compute the partial alignments scores

	-	A	G	A	T	T	C	C	A	T
-	0	0	0	0	0	0	0	0	0	0
A	0	1	1	1	1	1	1	1	1	1
G	0	1								
T	0									
C	0									
C	0									
C	0									
A	0									
T	0									

$$\nwarrow a_{i-1,j-1} + \sigma_{i,j} = 1 + \delta_{G,A} = 1$$

$$\leftarrow a_{i,j-1} = 0 + 0 = 0$$

$$\uparrow a_{i-1,j} = 1 + 0 = 1$$

1

Compute the partial alignments scores

	-	A	G	A	T	T	C	C	A	T
-	0	0	0	0	0	0	0	0	0	0
A	0	1	1	1	1	1	1	1	1	1
G	0	1	2							
T	0									
C	0									
C	0									
C	0									
A	0									
T	0									

$\nwarrow a_{i-1,j-1} + \sigma_{i,j} = 1 + \delta_{G,G} = 2$

Compute the partial alignments scores

	-	A	G	A	T	T	C	C	A	T
-	0	0	0	0	0	0	0	0	0	0
A	0	1	1	1	1	1	1	1	1	1
G	0	1	2							
T	0									
C	0									
C	0									
C	0									
A	0									
T	0									

↖ $a_{i-1,j-1} + \sigma_{i,j} = 1 + \delta_{G,G} = 2$

← $a_{i,j-1} = 1 + 0 = 1$

Compute the partial alignments scores

	-	A	G	A	T	T	C	C	A	T
-	0	0	0	0	0	0	0	0	0	0
A	0	1	1	1	1	1	1	1	1	1
G	0	1	2							
T	0									
C	0									
C	0									
C	0									
A	0									
T	0									

$$\nwarrow a_{i-1,j-1} + \sigma_{i,j} = 1 + \delta_{G,G} = 2$$

$$\leftarrow a_{i,j-1} = 1 + 0 = 1$$

$$\uparrow a_{i-1,j} = 1 + 0 = 1$$

Compute the partial alignments scores

	-	A	G	A	T	T	C	C	A	T
-	0	0	0	0	0	0	0	0	0	0
A	0	1	1	1	1	1	1	1	1	1
G	0	1	2							
T	0									
C	0									
C	0									
C	0									
A	0									
T	0									

$$\nwarrow a_{i-1,j-1} + \sigma_{i,j} = 1 + \delta_{G,G} = 2$$

$$\leftarrow a_{i,j-1} = 1 + 0 = 1$$

$$\uparrow a_{i-1,j} = 1 + 0 = 1$$

2

Compute the partial alignments scores

	-	A	G	A	T	T	C	C	A	T
-	0	0	0	0	0	0	0	0	0	0
A	0	1	1	1	1	1	1	1	1	1
G	0	1	2	2	2	2	2	2	2	2
T	0	1	2	2	3	3	3	3	3	3
C	0	1	2	2	3	3	4	4	4	4
C	0	1	2	2	3	3	4	5	5	5
C	0	1	2	2	3	3	4	5	5	5
A	0	1	2	3	3	3	4	5	6	6
T	0	1	2	3	4	4	4	5	6	7

Optimal alignment score

	-	A	G	A	T	T	C	C	A	T
-	0	0	0	0	0	0	0	0	0	0
A	0	1	1	1	1	1	1	1	1	1
G	0	1	2	2	2	2	2	2	2	2
T	0	1	2	2	3	3	3	3	3	3
C	0	1	2	2	3	3	4	4	4	4
C	0	1	2	2	3	3	4	5	5	5
C	0	1	2	2	3	3	4	5	5	5
A	0	1	2	3	3	3	4	5	6	6
T	0	1	2	3	4	4	4	5	6	7

Optimal alignment score

	-	A	G	A	T	T	C	C	A	T
-	0	0	0	0	0	0	0	0	0	0
A	0	1	1	1	1	1	1	1	1	1
G	0	1	2	2	2	2	2	2	2	2
T	0	1	2	2	3	3	3	3	3	3
C	0	1	2	2	3	3	4	4	4	4
C	0	1	2	2	3	3	4	5	5	5
C	0	1	2	2	3	3	4	5	5	5
A	0	1	2	3	3	3	4	5	6	6
T	0	1	2	3	4	4	4	5	6	7

And the optimal alignment?

Traceback

	-	A	G	A	T	T	C	C	A	T
-	.	.	.	.	.	.	.	.	.	.
A	.	↖	←	↖	←	←	←	←	↖	←
G	.	↑	↖	←	←	←	←	←	←	←
T	.	↖	↑	↖	↖	↖	←	←	←	↖
C	.	↑	↑	↖	↑	↖	↖	↖	←	←
C	.	↑	↑	↖	↑	↖	↖	↖	←	←
C	.	↑	↑	↖	↑	↖	↖	↖	↖	↖
A	.	↖	↑	↖	↑	↖	↑	↑	↖	←
T	.	↑	↑	↑	↖	↖	↑	↑	↑	↖

↖ $s_1[7]=T$ $s_2[8]=T$

Traceback

	-	A	G	A	T	T	C	C	A	T
-	.	.	.	.	.	.	.	.	.	.
A	.	↖	←	↖	←	←	←	←	↖	←
G	.	↑	↖	←	←	←	←	←	←	←
T	.	↖	↑	↖	↖	↖	←	←	←	↖
C	.	↑	↑	↖	↑	↖	↖	↖	←	←
C	.	↑	↑	↖	↑	↖	↖	↖	←	←
C	.	↑	↑	↖	↑	↖	↖	↖	↖	↖
A	.	↖	↑	↖	↑	↖	↑	↑	↖	←
T	.	↑	↑	↑	↖	↖	↑	↑	↑	↖

↖ $s_1[7]=T$ $s_2[8]=T$

T

T

Traceback

	-	A	G	A	T	T	C	C	A	T
-	.	.	.	.	.	.	.	.	.	.
A	.	↖	←	↖	←	←	←	←	↖	←
G	.	↑	↖	←	←	←	←	←	←	←
T	.	↖	↑	↖	↖	↖	←	←	←	↖
C	.	↑	↑	↖	↑	↖	↖	↖	←	←
C	.	↑	↑	↖	↑	↖	↖	↖	←	←
C	.	↑	↑	↖	↑	↖	↖	↖	↖	↖
A	.	↖	↑	↖	↑	↖	↑	↑	↖	←
T	.	↑	↑	↑	↖	↖	↑	↑	↑	↖

↖ $s_1[6]=A$ $s_2[7]=A$

Traceback

	-	A	G	A	T	T	C	C	A	T
-	.	.	.	.	.	.	.	.	.	.
A	.	↖	←	↖	←	←	←	←	↖	←
G	.	↑	↖	←	←	←	←	←	←	←
T	.	↖	↑	↖	↖	↖	←	←	←	↖
C	.	↑	↑	↖	↑	↖	↖	↖	←	←
C	.	↑	↑	↖	↑	↖	↖	↖	←	←
C	.	↑	↑	↖	↑	↖	↖	↖	↖	↖
A	.	↖	↑	↖	↑	↖	↑	↑	↖	←
T	.	↑	↑	↑	↖	↖	↑	↑	↑	↖

↖ $s_1[6]=A$ $s_2[7]=A$

TA

TA

Traceback

	-	A	G	A	T	T	C	C	A	T
-	.	.	.	.	.	.	.	.	.	.
A	.	↖	←	↖	←	←	←	←	↖	←
G	.	↑	↖	←	←	←	←	←	←	←
T	.	↖	↑	↖	↖	↖	←	←	←	↖
C	.	↑	↑	↖	↑	↖	↖	↖	←	←
C	.	↑	↑	↖	↑	↖	↖	↖	←	←
C	.	↑	↑	↖	↑	↖	↖	↖	↖	↖
A	.	↖	↑	↖	↑	↖	↑	↑	↖	←
T	.	↑	↑	↑	↖	↖	↑	↑	↑	↖

↖ $s_1[5]=C$ $s_2[6]=C$

Traceback

	-	A	G	A	T	T	C	C	A	T
-	.	.	.	.	.	.	.	.	.	.
A	.	↖	←	↖	←	←	←	←	↖	←
G	.	↑	↖	←	←	←	←	←	←	←
T	.	↖	↑	↖	↖	↖	←	←	←	↖
C	.	↑	↑	↖	↑	↖	↖	↖	←	←
C	.	↑	↑	↖	↑	↖	↖	↖	←	←
C	.	↑	↑	↖	↑	↖	↖	↖	↖	↖
A	.	↖	↑	↖	↑	↖	↑	↑	↖	←
T	.	↑	↑	↑	↖	↖	↑	↑	↑	↖

↖ $s_1[5]=C$ $s_2[6]=C$

TAC

TAC

Traceback

	-	A	G	A	T	T	C	C	A	T
-	.	.	.	.	.	.	.	.	.	.
A	.	↖	←	↖	←	←	←	←	↖	←
G	.	↑	↖	←	←	←	←	←	←	←
T	.	↖	↑	↖	↖	↖	←	←	←	↖
C	.	↑	↑	↖	↑	↖	↖	↖	←	←
C	.	↑	↑	↖	↑	↖	↖	↖	←	←
C	.	↑	↑	↖	↑	↖	↖	↖	↖	↖
A	.	↖	↑	↖	↑	↖	↑	↑	↖	←
T	.	↑	↑	↑	↖	↖	↑	↑	↑	↖

↖ $s_1[4]=C$ $s_2[5]=C$

Traceback

	-	A	G	A	T	T	C	C	A	T
-	.	.	.	.	.	.	.	.	.	.
A	.	↖	←	↖	←	←	←	←	↖	←
G	.	↑	↖	←	←	←	←	←	←	←
T	.	↖	↑	↖	↖	↖	←	←	←	↖
C	.	↑	↑	↖	↑	↖	↖	↖	←	←
C	.	↑	↑	↖	↑	↖	↖	↖	←	←
C	.	↑	↑	↖	↑	↖	↖	↖	↖	↖
A	.	↖	↑	↖	↑	↖	↑	↑	↖	←
T	.	↑	↑	↑	↖	↖	↑	↑	↑	↖

↖ $s_1[4]=C$ $s_2[5]=C$

TACC

TACC

Traceback

	-	A	G	A	T	T	C	C	A	T
-	.	.	.	.	.	.	.	.	.	.
A	.	↖	←	↖	←	←	←	←	↖	←
G	.	↑	↖	←	←	←	←	←	←	←
T	.	↖	↑	↖	↖	↖	←	←	←	↖
C	.	↑	↑	↖	↑	↖	↖	↖	←	←
C	.	↑	↑	↖	↑	↖	↖	↖	←	←
A	.	↖	↑	↖	↑	↖	↑	↑	↖	←
T	.	↑	↑	↑	↖	↖	↑	↑	↑	↖

↖ $s_1[3]=C$ $s_2[4]=T$

Traceback

	-	A	G	A	T	T	C	C	A	T
-	.	.	.	.	.	.	.	.	.	.
A	.	↖	←	↖	←	←	←	←	↖	←
G	.	↑	↖	←	←	←	←	←	←	←
T	.	↖	↑	↖	↖	↖	←	←	←	↖
C	.	↑	↑	↖	↑	↖	↖	↖	←	←
C	.	↑	↑	↖	↑	↖	↖	↖	←	←
A	.	↖	↑	↖	↑	↖	↑	↑	↖	←
T	.	↑	↑	↑	↖	↖	↑	↑	↑	↖

↖ $s_1[3]=C$ $s_2[4]=T$

TACCC

TACCT

Traceback

	-	A	G	A	T	T	C	C	A	T
-	.	.	.	.	.	.	.	.	.	.
A	.	↖	←	↖	←	←	←	←	↖	←
G	.	↑	↖	←	←	←	←	←	←	←
T	.	↖	↑	↖	↖	↖	←	←	←	↖
C	.	↑	↑	↖	↑	↖	↖	↖	←	←
C	.	↑	↑	↖	↑	↖	↖	↖	←	←
C	.	↑	↑	↖	↑	↖	↖	↖	↖	↖
A	.	↖	↑	↖	↑	↖	↑	↑	↖	←
T	.	↑	↑	↑	↖	↖	↑	↑	↑	↖

↖ $s_1[2]=T$ $s_2[3]=T$

Traceback

	-	A	G	A	T	T	C	C	A	T
-	.	.	.	.	.	.	.	.	.	.
A	.	↖	←	↖	←	←	←	←	↖	←
G	.	↑	↖	←	←	←	←	←	←	←
T	.	↖	↑	↖	↖	↖	←	←	←	↖
C	.	↑	↑	↖	↑	↖	↖	↖	←	←
C	.	↑	↑	↖	↑	↖	↖	↖	←	←
C	.	↑	↑	↖	↑	↖	↖	↖	↖	↖
A	.	↖	↑	↖	↑	↖	↑	↑	↖	←
T	.	↑	↑	↑	↖	↖	↑	↑	↑	↖

↖ $s_1[2]=T$ $s_2[3]=T$

TACCCT

TACCTT

Traceback

	-	A	G	A	T	T	C	C	A	T
-	.	.	.	.	.	.	.	.	.	.
A	.	↖	←	↖	←	←	←	←	↖	←
G	.	↑	↖	←	←	←	←	←	←	←
T	.	↖	↑	↖	↖	↖	←	←	←	↖
C	.	↑	↑	↖	↑	↖	↖	↖	←	←
C	.	↑	↑	↖	↑	↖	↖	↖	←	←
C	.	↑	↑	↖	↑	↖	↖	↖	↖	↖
A	.	↖	↑	↖	↑	↖	↑	↑	↖	←
T	.	↑	↑	↑	↖	↖	↑	↑	↑	↖

← - $s_2[2]=A$

Traceback

	-	A	G	A	T	T	C	C	A	T
-	.	.	.	.	.	.	.	.	.	.
A	.	↖	←	↖	←	←	←	←	↖	←
G	.	↑	↖	←	←	←	←	←	←	←
T	.	↖	↑	↖	↖	↖	←	←	←	↖
C	.	↑	↑	↖	↑	↖	↖	↖	←	←
C	.	↑	↑	↖	↑	↖	↖	↖	←	←
C	.	↑	↑	↖	↑	↖	↖	↖	↖	↖
A	.	↖	↑	↖	↑	↖	↑	↑	↖	←
T	.	↑	↑	↑	↖	↖	↑	↑	↑	↖

← - $s_2[2]=A$

TACCCT-

TACCTTA

Traceback

	-	A	G	A	T	T	C	C	A	T
-	.	.	.	.	.	.	.	.	.	.
A	.	↖	←	↖	←	←	←	←	↖	←
G	.	↑	↖	←	←	←	←	←	←	←
T	.	↖	↑	↖	↖	↖	←	←	←	↖
C	.	↑	↑	↖	↑	↖	↖	↖	←	←
C	.	↑	↑	↖	↑	↖	↖	↖	←	←
C	.	↑	↑	↖	↑	↖	↖	↖	↖	↖
A	.	↖	↑	↖	↑	↖	↑	↑	↖	←
T	.	↑	↑	↑	↖	↖	↑	↑	↑	↖

↖ $s_1[1]=G$ $s_2[1]=G$

Traceback

	-	A	G	A	T	T	C	C	A	T
-	.	.	.	.	.	.	.	.	.	.
A	.	↖	←	↖	←	←	←	←	↖	←
G	.	↑	↖	←	←	←	←	←	←	←
T	.	↖	↑	↖	↖	↖	←	←	←	↖
C	.	↑	↑	↖	↑	↖	↖	↖	←	←
C	.	↑	↑	↖	↑	↖	↖	↖	←	←
C	.	↑	↑	↖	↑	↖	↖	↖	↖	↖
A	.	↖	↑	↖	↑	↖	↑	↑	↖	←
T	.	↑	↑	↑	↖	↖	↑	↑	↑	↖

↖ $s_1[1]=G$ $s_2[1]=G$

TACCCT-G

TACCTTAG

Traceback

	-	A	G	A	T	T	C	C	A	T
-	.	.	.	.	.	.	.	.	.	.
A	.	↖	←	↖	←	←	←	←	↖	←
G	.	↑	↖	←	←	←	←	←	←	←
T	.	↖	↑	↖	↖	↖	←	←	←	↖
C	.	↑	↑	↖	↑	↖	↖	↖	←	←
C	.	↑	↑	↖	↑	↖	↖	↖	←	←
C	.	↑	↑	↖	↑	↖	↖	↖	↖	↖
A	.	↖	↑	↖	↑	↖	↑	↑	↖	←
T	.	↑	↑	↑	↖	↖	↑	↑	↑	↖

↖ $s_1[0]=A$ $s_2[0]=A$

Traceback

	-	A	G	A	T	T	C	C	A	T
-	.	.	.	.	.	.	.	.	.	.
A	.	↖	←	↖	←	←	←	←	↖	←
G	.	↑	↖	←	←	←	←	←	←	←
T	.	↖	↑	↖	↖	↖	←	←	←	↖
C	.	↑	↑	↖	↑	↖	↖	↖	←	←
C	.	↑	↑	↖	↑	↖	↖	↖	←	←
C	.	↑	↑	↖	↑	↖	↖	↖	↖	↖
A	.	↖	↑	↖	↑	↖	↑	↑	↖	←
T	.	↑	↑	↑	↖	↖	↑	↑	↑	↖

↖ $s_1[0]=A$ $s_2[0]=A$

TACCCT-GA

TACCTTAGA

Traceback

	-	A	G	A	T	T	C	C	A	T
-	.	.	.	.	.	.	.	.	.	.
A	.	↖	←	↖	←	←	←	←	↖	←
G	.	↑	↖	←	←	←	←	←	←	←
T	.	↖	↑	↖	↖	↖	←	←	←	↖
C	.	↑	↑	↖	↑	↖	↖	↖	←	←
C	.	↑	↑	↖	↑	↖	↖	↖	←	←
C	.	↑	↑	↖	↑	↖	↖	↖	↖	↖
A	.	↖	↑	↖	↑	↖	↑	↑	↖	←
T	.	↑	↑	↑	↖	↖	↑	↑	↑	↖

HALT! - -

Traceback

	-	A	G	A	T	T	C	C	A	T
-	.	.	.	.	.	.	.	.	.	.
A	.	↖	←	↖	←	←	←	←	↖	←
G	.	↑	↖	←	←	←	←	←	←	←
T	.	↖	↑	↖	↖	↖	←	←	←	↖
C	.	↑	↑	↖	↑	↖	↖	↖	←	←
C	.	↑	↑	↖	↑	↖	↖	↖	←	←
C	.	↑	↑	↖	↑	↖	↖	↖	↖	↖
A	.	↖	↑	↖	↑	↖	↑	↑	↖	←
T	.	↑	↑	↑	↖	↖	↑	↑	↑	↖

HALT! - -

TACCCT-GA

TACCTTAGA

Reverse the traceback

original strings:

AGTCCCAT

AGATTCCAT

traceback strings:

TACCCT-GA

TACCTTAGA

Reverse the traceback

original strings:

AGTCCCAT

AGATTCCAT

traceback strings:

TACCCT-GA

TACCTTAGA

aligned strings:

AG-TCCCAT

AGATTCCAT